

Some devices won't be able to accommodate the entire CPython inside their memory. In these cases a modified or adapted version of Python needs to be used. Examples of such implementations are PyMite and Tiny Python. These versions of Python are so heavily reengineered that you can even get away with calling them a different version of Python. PyMite for eg., is a flyweight Python interpreter which can execute even on 8-bit microcontrollers with resources as limited as 64K Flash and just 4K of RAM. PyMite supports a subset of the Python 2.5 syntax and can execute a subset of the Python 2.5 bytecode.

Other Languages

Apart from these languages there are a lot of other languages which are used by some for programming embedded systems. The more important ones that we'll discuss are Forth, Ada and Basic.

Forth is a stack-based language and a software development environment. The language much like Python is useless without its interpreter. Working with Forth is different compared to other programming languages, it works in a more 'hacky' way. You program with Forth as if you're making a complex shell script. You write a little, test a little and repeat the process till you get the result. Forth is an interactive language so it makes it very easy to interact with the target system. Forth's extensibility is also worth a mention. Forth lets the developer extend and control the compiler itself, something which we cannot do in C. Forth is currently used in is currently used in boot loaders such as Open Firmware, space applications and other embedded systems.

Ada is other languages that has achieved some measure of success in the embedded system market. Ada was conceived to work exclusively on embedded systems. Developed by DoD, the focus when developing the language was the safety and reliability of the systems designed using Ada. Ada compiler go through rigorous checkings against the exhaustive ACATS (Ada Conformity Assessment Test Suite), because of these safety features it finds extensive usage in military, aviation and air traffic control, commercial rockets, satellites and other space systems.

And finally BASIC. You may be surprised to find this language here, as it has always considered to be something of a toy language to learn programming. But Basic finds serious usage in embedded system. Basic's simplicity is its greatest strength – it is very easy to learn. It runs at roughly 95,000 instructions per second. A lot of boards come